

SaaS Growth Economics

Diagnosing recurring-revenue growth, retention, and acquisition efficiency

Self-directed portfolio project using synthetic data

BUSINESS QUESTION

How should Finance evaluate whether recurring-revenue growth is durable when ARR, retention, customer mix, and acquisition activity are moving in different directions?

ANALYSIS WINDOW Apr 2024 – Sep 2025	AS-OF DATE 2025-09-30	CUSTOMERS 100 synthetic logos	ARR BASIS Run-rate (MRR × 12)
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GROWTH DIAGNOSIS

ARR peaked at \$73,560 in Jul 2025, then declined approximately 19% to \$59,400 by Sep 2025. There was no true expansion engine in this fixture.

\$73,560
peak run-rate ARR in Jul 2025

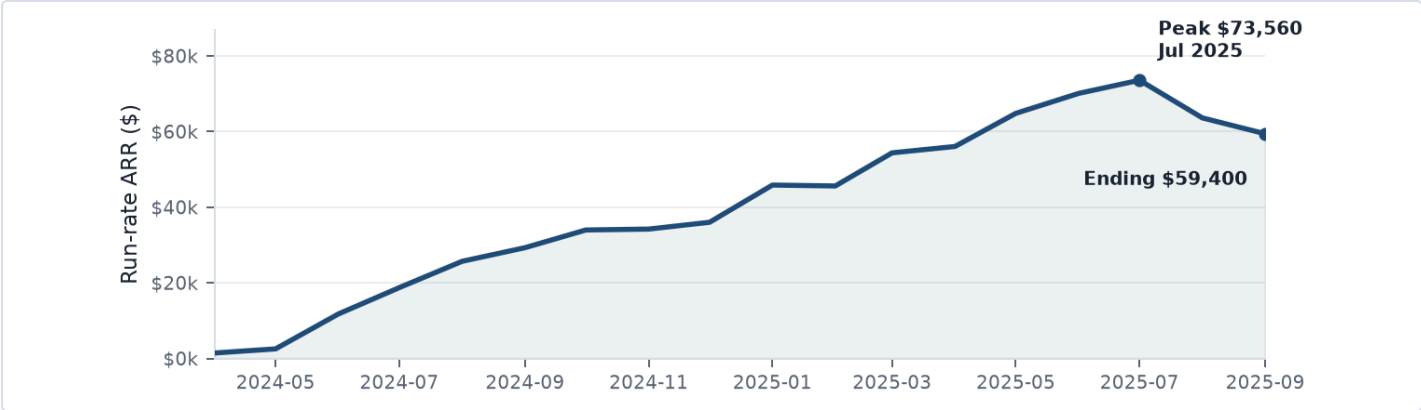


Figure 1. Run-rate ARR. Peak \$73,560 in Jul 2025; ending \$59,400 (\$4,950 MRR).



Figure 2. ARR bridge, April–September 2025. Expansion and contraction are \$0 because the fixture contains no upgrades or downgrades. Reactivation is shown separately from expansion.

FP&A diagnosis: Growth depended on new-logo acquisition, with reactivation providing a smaller offset and no true expansion. From April through September 2025, \$32.9k of new ARR was offset by \$32.9k of churn ARR; \$5.0k of reactivation was the only net bridge contribution. This is an acquisition-dependent growth engine with limited recurring-revenue durability.

Retention never generated net expansion

Revenue quality of the beginning recurring-revenue base

NRR and GRR are identical in this fixture. There were no true upgrades and no contractions, so movement inside the beginning base is churn only. New logos and reactivations are excluded from both rates.

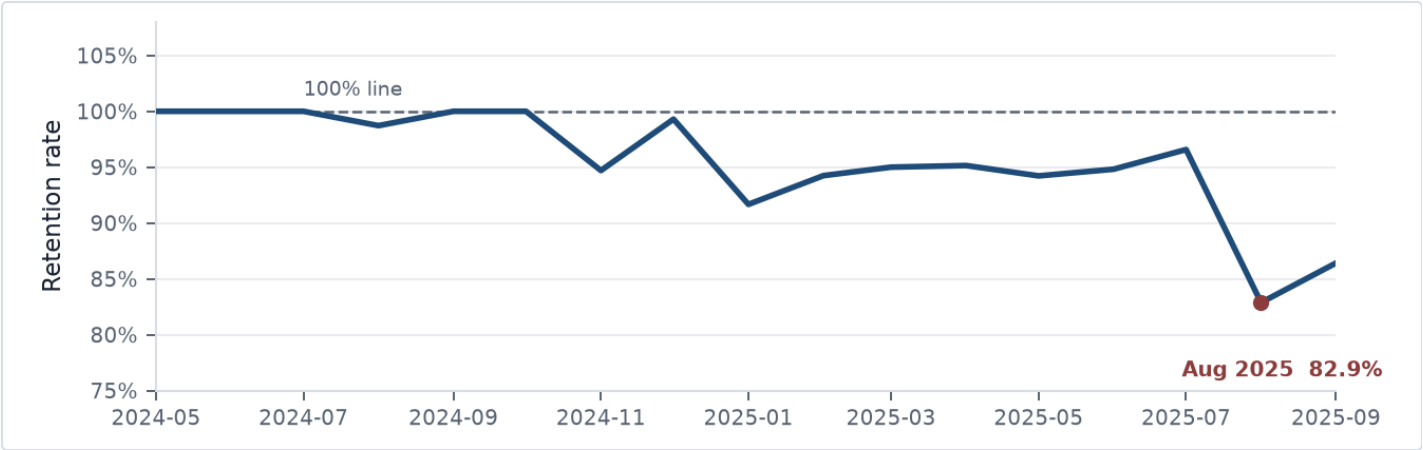


Figure 3. Monthly NRR / GRR. Mean 95.5%. August 2025: 82.9%. Months with NRR above 100%: 0.

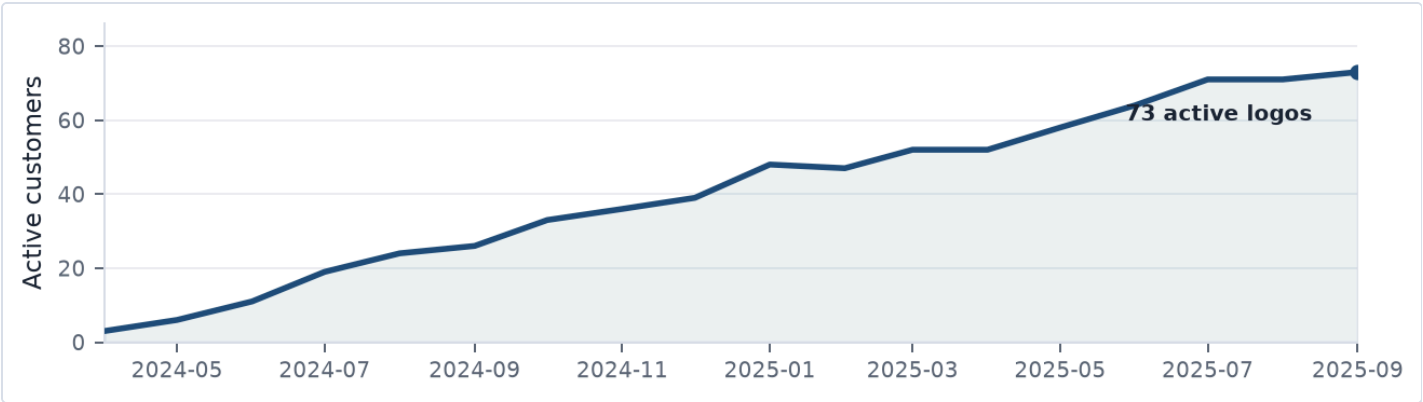


Figure 4. Active customers rose to 73 by Sep 2025 even as ARR fell from the July peak. Logo growth did not prevent a revenue decline.

Risks

- Mean NRR of 95.5% means the existing base shrinks unless new logos or reactivation replace leakage.
- August NRR of 82.9% shows how quickly the beginning base can compress when churn spikes. Slower new-logo contribution then leaves less offset at the company level.
- A model dependent on acquisition must continually replace recurring-revenue leakage. That is a less durable plan than expansion-led growth.

Opportunities

- A real expansion / upsell motion would be the cleanest way to lift NRR above GRR, if product and customer behavior support it.
- Finance should segment churn by channel, customer segment, cohort, and tenure before increasing acquisition spend.
- Reactivation helped, but it is not a substitute for retaining the beginning base or building expansion.

Interpretation: These are analytical recommendations for how Finance should review the book, not actions that were implemented. Signup-cohort dollar retention can exceed 100% here because some subscriptions start after signup month and some logos later reactivate; it is not NRR and is not shown as a heatmap for that reason.

Acquisition efficiency requires tighter investment discipline

CAC payback uses monthly contribution margin: $CAC \div (ARPU \times 80\% \text{ gross margin})$

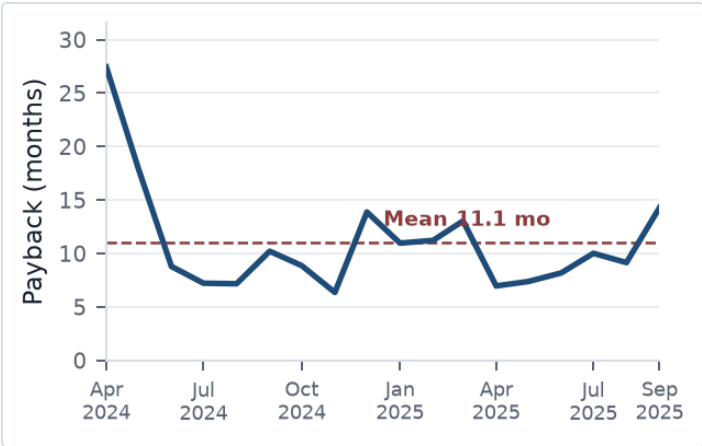


Figure 5. Corrected CAC payback. Mean 11.1 months. Assigned customer CAC, not GL marketing spend.

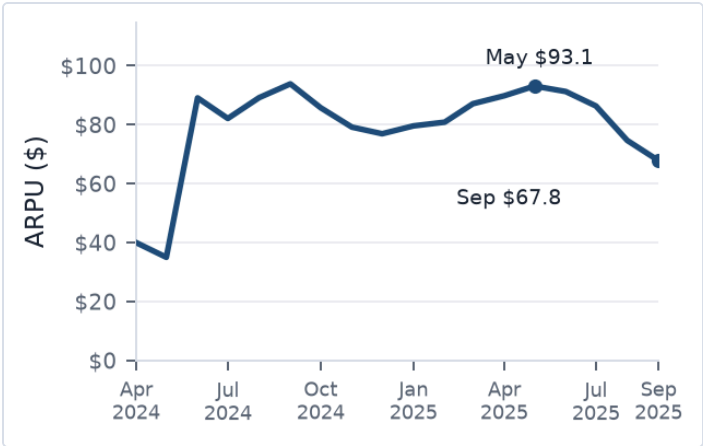


Figure 6. Blended ARPU fell from \$93.1 in May 2025 to \$67.8 in Sep 2025, reducing contribution dollars available to recover CAC.

Channel snapshot (interpret with small-sample caution)

CHANNEL	NEW LOGOS	AVG ASSIGNED CAC	SEP ARPU	PAYBACK PROXY
Paid	39	\$931	\$61	19.2 mo
Organic	31	\$213	\$94	2.8 mo
Partner	20	\$732	\$58	15.9 mo
Outbound	10	\$560	\$37	18.8 mo

Payback proxy = average assigned CAC ÷ (September channel ARPU × 80% gross margin). Useful for mix discussion, not a vintage payback curve. Outbound and Partner samples are small.

Management actions for Finance to challenge

- 1. Separate the forecast into beginning-base retention, new-logo acquisition, reactivation, and expansion so growth is not treated as a single rate.
- 2. Investigate churn concentration by segment, channel, and tenure before increasing acquisition spend.
- 3. Establish an explicit expansion / upsell driver rather than relying on new logos to replace leakage.
- 4. Use CAC payback thresholds that update as ARPU and retention change; do not freeze an 11-month average as a permanent hurdle.

What this book is telling Finance

- Headline ARR growth through mid-2025 masked a missing expansion engine.
- Falling ARPU lengthens effective payback even when monthly CAC is not rising.
- Paid acquisition is the largest logo source and has the highest payback proxy in this mix.

Limitations

- Synthetic data; results do not describe a real company.
- Frozen window through 2025-09-30.
- Assigned customer CAC, not GL marketing-spend allocation.
- No true upgrade/downgrade activity in the fixture.
- No causal inference and no production deployment.